

G5_Task

Hello,

Please read the following carefully:

As a Software Engineering research team at the Polytechnique Montréal (Canada), we are interested in understanding developers' assessment of coding task difficulty. You will be provided with a series of 12 coding questions. Questions are divided into two parts:

- First, you will be given three instructions representing coding tasks and you will be asked to write, in a few words, what is the underlying task represented by all these instructions. The instructions represent the same tasks, but they vary in wording and information they give. In a nutshell, you have to distill the essence of the task and write it in a few words.

- In a second time, you will be asked to assess how difficult implementing this **task** would be. Note we ask the assessment based on the task you will have worded in the previous part, not based on any of the instructions presented earlier. First, you will be asked your subjective judgment on a scale from Very Easy to Very Difficult. Then, to make judging the difficulty more relatable, we adopt a similar approach as what is used in agile development and planning poker, i.e. you will be asked to evaluate how long it would take to implement a task (in minutes here). Consider in your assessment the knowledge needed to implement the task, the external information you would need to use, whether you would need the assistance of a LLM such as ChatGPT (and so the time it would take to debug the obtained code) etc. Finally, you will have to explain in a few words why you gave such difficulty ratings.

To give you an idea of the range of tasks you will need to assess, the first section of this questionnaire will be two examples we devised to illustrate the questionnaire.

Completing the task will take about 30 minutes.

Note also that all examples are written using Python 3.10 syntax.

Please make sure to answer all the questions to the best of your ability.

Please note that your identity and personal data will not be disclosed, while we plan to use the aggregated results and anonymized responses as part of a scientific publication. If you have any questions about the questionnaire or our research, please do not hesitate to contact us.

By proceeding, you consent to take part in the study under the conditions described previously.

Thank you,

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* Indique une question obligatoire

1. Please enter your prolific ID : *

Example of tasks

As we ask of you some estimation of difficulty and the tasks dealt in this questionnaire can be a bit different from what you can be used to as a developer, we give you examples of some tasks along with some assessments.

Note that this assessment is **our assessment**. You may have a different opinion based on your background. The idea of those examples is to give an extreme range of what type of tasks you can expect in the questionnaire.

As a reminder: You will first be given three instructions and asked to explain what the task is about in a few words. Then, you will have to rate the difficulty of the **task** you just described (not accounting for particular instruction). Rating the difficulty is done in two steps: subjective difficulty and approximate time needed. Finally, you will be asked to say why you gave such rating.

Example 1

Instruction 1

```
1 def subtract(c1, c2):  
2     """  
3     Deliver the result of subtracting two complex numbers,  
4     specified as 'c1' and 'c2', as a complex output.  
5     :param c1: The first complex number, complex.  
6     :param c2: The second complex number, complex.  
7     :return: The difference of the two complex numbers, complex.  
8     """
```

Instruction 2

```
1 def subtract(c1, c2):
2     """
3     The method receives two complex numbers 'c1' and 'c2', from which it
4     subtracts 'c2.real' from 'c1.real' to determine the real portion, and
5     subtracts 'c2.imag' from 'c1.imag' to determine the imaginary portion.
6     It constructs and returns a new complex number using these results
7     with the 'complex()' function.
8     :param c1: The first complex number, complex.
9     :param c2: The second complex number, complex.
10    :return: The difference of the two complex numbers, complex.
11    """
```

Instruction 3

```
1 def subtract(c1, c2):
2     """
3     Subtracts two complex numbers.
4     :param c1: The first complex number, complex.
5     :param c2: The second complex number, complex.
6     :return: The difference of the two complex numbers, complex.
7     >>> complexCalculator = ComplexCalculator()
8     >>> complexCalculator.subtract(1+2j, 3+4j)
9     (-2-2j)
10    """
```

Answer

The task encapsulated by the **subtract**

function

is to compute the subtraction between two complex numbers.

This task has a subjective difficulty of **Very Easy** as assessed by us and would take about **1 min**

to implement. There is no particular difficulty, the task is very straightforward without any corner case, it just requires to know what a complex is, and the standard Python library for it.

Example 2

Context

```
1 import numpy as np
2 class MetricsCalculator2:
3     """
4     The class provides to calculate Mean Reciprocal Rank (MRR) and Mean Average Precision (MAP)
5     based on input data, where MRR measures the ranking quality and MAP measures the average precision.
6     """
7
8     def __init__(self):
9         pass
10
11     @staticmethod
12     def map(data):
13         pass
```

Instruction 1

```
1 def mrr(data):
2     """
3     Compute the MRR of the input data. MRR is a widely used evaluation index.
4     It is the mean of reciprocal rank.
5     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
6     In each tuple (actual result,ground truth num),ground truth num is the total ground num.
7     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
8     1 stands for a correct answer, 0 stands for a wrong answer.
9     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
10     average recall on all list. The second return value is a list of precision for each input.
11     >>> MetricsCalculator2.mrr([([1, 0, 1, 0], 4)])
12     >>> MetricsCalculator2.mrr([([1, 0, 1, 0], 4), ([0, 1, 0, 1], 4)])
13     1.0, [1.0]
14     0.75, [1.0, 0.5]
15
16     """
```

Instruction 2

```
1 def mrr(data):
2     """
3     Assess the Mean Reciprocal Rank (MRR) from the input 'data'.
4     MRR assesses the average of reciprocal ranks of outcomes.
5     The input structure 'data' must be a tuple consisting of a list of
6     binary values and its total count or a list of such tuples.
7     Each binary value (0 or 1) represents the correctness of an answer.
8     If 'data' is a tuple, the function returns its mean reciprocal rank.
9     If 'data' is a list, the function returns the overall average MRR along
10     with a list showing reciprocal ranks for each component tuple.
11
12     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
13     In each tuple (actual result,ground truth num),ground truth num is the total ground num.
14     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
15     1 stands for a correct answer, 0 stands for a wrong answer.
16     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
17     average recall on all list. The second return value is a list of precision for each input.
18
19     """
```

Instruction 3

```
1 def mrr(data):
2     """
3     Evaluate the Mean Reciprocal Rank (MRR) from 'data'. This metric calculates the mean of reciprocal
4     ranks for correct answers. Input 'data' could be either a tuple containing binary correctness indicators
5     and the aggregate count of answers, or multiple tuples of this type in a list. For each tuple, the function
6     measures reciprocal ranks by multiplying the binary indicators by reciprocals of their positions, finding the
7     earliest nonzero value to establish the rank of the first correct response. If the input 'data' is a tuple,
8     the MRR calculated for this specific tuple is returned; if it's a list, the function summarizes by providing
9     both the average MRR and a list of individual MRR values for each tuple.
10
11     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
12     In each tuple (actual result, ground truth num), ground truth num is the total ground num.
13     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5), ([1,0,...],6), ([0,0,...],5)].
14     1 stands for a correct answer, 0 stands for a wrong answer.
15     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
16     average recall on all list. The second return value is a list of precision for each input.
17
18     """
```

Answer

The task encapsulated by the **mrr** function is to compute the Mean Reciprocal Rank of binary tuples data.

This task has a subjective difficulty of **Very Hard** as assessed by us and would take about **40 min**

to implement. The difficulty mainly branches from dealing with the tuple data and how the formula is processed accordingly (with a few corner cases). This can take additional time if the Mean Reciprocal Rank is not known.

Question 1:

Below are instructions for implementing the function.

Instruction 1

```
1 def max_element(l: list):
2     """
3     Construct a function named 'max_element' which is
4     designed to take a number list as an argument.
5     This function aims to retrieve and return the maximum
6     number from the list using a built-in function,
7     bypassing any manual comparisons or iterations.
8     """
```

Instruction 2

```
1 def max_element(l: list):
2     """
3     Return maximum element in the list.
4     >>> max_element([1, 2, 3])
5     3
6     >>> max_element([5, 3, -5, 2, -3, 3, 9, 0, 123, 1, -10])
7     123
8     """
```

Instruction 3

```
1 def max_element(l: list):
2     """
3     Develop a function 'max_element' that receives a list of numbers.
4     Its role is to identify and return the highest value in the list
5     by leveraging a built-in function, thus eliminating the need for
6     manual element comparisons or iterations.
7     """
```

2. In a few words, what is the task this function aims to implement? *

3. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

4. How would you rate the difficulty of this task, in minutes it would take to implement?

*

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

5. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 2:

Below are instructions for implementing the function:

Context

```
1 class JobMarketplace:
2     """
3     This is a class that provides functionalities to publish positions,
4     remove positions, submit resumes, withdraw resumes, search for positions,
5     and obtain candidate information.
6     """
7
8     def __init__(self):
9         self.job_listings = []
10        self.resumes = []
11
12    def post_job(self, job_title, company, requirements):
13        pass
14
15    def remove_job(self, job):
16        pass
17
18    def submit_resume(self, name, skills, experience):
19        pass
20
21    def search_jobs(self, criteria):
22        pass
23
24    def get_job_applicants(self, job):
25        pass
26
27
28
```

Instruction 1

```
1 def withdraw_resume(self, resume):
2     """
3     Utilized for extracting resumes, this function erases the
4     information of a selected resume from the 'self.resumes' list.
5     It initiates by detecting the targeted resume to be extracted.
6     Subsequent to its detection, the function executes the removal
7     of the chosen resume from the 'self.resumes' list, thereby
8     updating the list to mirror the extraction of the resume entry.
9     :param resume: The resume information to be removed,dict.
10    :return: None
11    """
```

Instruction 2

```
1 def withdraw_resume(self, resume):
2     """
3     This function is used to withdraw resumes, and remove the
4     resume information from the resumes list.
5     :param resume: The resume information to be removed, dict.
6     :return: None
7     >>> jobMarketplace = JobMarketplace()
8     >>> jobMarketplace.resumes = [{"name": "Tom", "skills":
9         ['skill1', 'skill2'], "experience": "experience"}]
10    >>> jobMarketplace.withdraw_resume(jobMarketplace.resumes[0])
11    >>> jobMarketplace.resumes
12    []
13    """
```

Instruction 3

```
1 def withdraw_resume(self, resume):
2     """
3     This function serves to extract and erase resume data
4     from the 'self.resumes' collection.
5     :param resume: The resume information to be removed, dict.
6     :return: None
7     """
```

6. In a few words, what is the task this function aims to implement? *

7. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

☐ Very Easy

☐ Easy

☐ Not Easy Nor Difficult

☐ Difficult

☐ Very Difficult

8. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

9. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 3:

Below are instructions for implementing the function:

Instruction 1

```
1 def closest_integer(value):
2     """
3     Create a function that takes a value (string) representing a number
4     and returns the closest integer to it. If the number is equidistant
5     from two integers, round it away from zero.
6
7     Examples
8     >>> closest_integer("10")
9     10
10    >>> closest_integer("15.3")
11    15
12
13    Note:
14    Rounding away from zero means that if the given number is equidistant
15    from two integers, the one you should return is the one that is the
16    farthest from zero. For example closest_integer("14.5") should
17    return 15 and closest_integer("-14.5") should return -15.
18    """
```

Instruction 2

```
1 def closest_integer(value):
2     """
3     Define a function 'closest_integer' that takes a single string input 'value',
4     denoting a number. Initially, this string is converted to a float.
5     A nested function 'rounding()', taking 'val' as an input, checks whether
6     the fractional difference from the whole number is not 0.5
7     (using 'abs(val - int(val)) != 0.5').
8     If this is the case, it rounds 'val' with 'round(val)'.
9     If the fraction is 0.5 and 'val' is above zero, it returns
10    'int(val) + 1'. If below zero, it returns 'int(val) - 1'.
11    The outer function then returns what 'rounding()' returns after converting
12    'value' to a float.
13    """
```

Instruction 3

```
1 def closest_integer(value):
2     """
3     Write a function named 'closest_integer' that takes a string
4     input representing a numeric value. The function first converts
5     this string into a float. It then determines the closest integer
6     to this float value. If the decimal part of the float is exactly 0.5,
7     indicating that it is equidistant between two integers, the function
8     rounds the number away from zero. Otherwise, it rounds to the nearest
9     integer using standard rounding rules. Finally, the function returns
10    the integer.
11    """
```

10. In a few words, what is the task this function aims to implement? *

11. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

12. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

13. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 4:

Below are instructions for implementing the function:

Context

```
1 import socket
2 class IpUtil:
3     """
4     This is a class as tool for ip that can be used to obtain the local IP address,
5     validate its validity, and also provides the functionality to retrieve the
6     corresponding hostname.
7     """
8
9     def is_valid_ipv4(ip_address):
10         pass
11
12     @staticmethod
13     def is_valid_ipv6(ip_address):
14         pass
15
16     @staticmethod
17     def get_hostname(ip_address):
18
19
```

Instruction 1

```
1 def get_hostname(ip_address):
2     """
3     Get the hostname associated with the given IP address ""ip_address"".
4     If successful, return the hostname as a string; if the IP address
5     cannot be resolved, return ""None"". This function attempts to resolve
6     the IP address to a hostname by using a method that interacts with the
7     network's domain name system. If a hostname is found, it is returned.
8     Otherwise, if an error occurs during the resolution process, ""None""
9     is returned to indicate that the hostname could not be found.
10     :param ip_address:string, the IP address to get the hostname for
11     :return: string, the hostname associated with the IP address
12     """
```

Instruction 2

```
1 def get_hostname(ip_address):
2     """
3     Get the hostname associated with the given IP address.
4     :param ip_address:string, the IP address to get the hostname for
5     :return: string, the hostname associated with the IP address
6     >>> IpUtil.get_hostname('110.242.68.3')
7     'www.baidu.com'
8     >>> IpUtil.get_hostname('10.0.0.1')
9     """
```

Instruction 3

```
1 def get_hostname(ip_address):  
2     """  
3     Determine the hostname related to the indicated IP address 'ip_address'.  
4     If the hostname is available, return it; if not, yield 'None'.  
5     :param ip_address:string, the IP address to get the hostname for  
6     :return: string, the hostname associated with the IP address"  
7     """
```

14. In a few words, what is the task this function aims to implement? *

15. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

16. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

17. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 5:

Below are instructions for implementing the function:

Instruction 1

```
1 def correct_bracketing(brackets: str):
2     """
3     Write a function named 'correct_bracketing' that takes a string
4     comprised solely of the characters '<' and '>'.
5     The function should assess whether the string correctly brackets
6     each opening '<' with a corresponding closing '>'. It operates by
7     iterating through the string, maintaining a count that increases
8     with each '<' and decreases with each '>'. If at any point this
9     count becomes negative, the function immediately returns False,
10    indicating an unmatched closing bracket. The function returns
11    True only if the count is zero after processing the entire string,
12    ensuring all brackets are correctly paired.
13    """
```

Instruction 2

```
1 def correct_bracketing(brackets: str):
2     """
3     brackets is a string of "<" and ">".
4     return True if every opening bracket has a
5     corresponding closing bracket.
6
7     >>> correct_bracketing("<")
8     False
9     >>> correct_bracketing("<>")
10    True
11    >>> correct_bracketing("<<>>")
12    True
13    >>> correct_bracketing("><>")
14    False
15    """
```

Instruction 3

```
1 def correct_bracketing(brackets: str):
2     """
3     Construct a function named 'correct_bracketing' that takes an
4     input string composed entirely of the characters '<' and '>'.
5     This function verifies proper bracketing by incrementing a counter
6     for each '<' and decrementing it for each '>', throughout the string.
7     Should the counter fall below zero at any stage, the function is to
8     return False to denote that a closing bracket does not match any
9     preceding opening bracket. If the counter resets to zero after scanning
10    the complete string, it returns True to signify that every bracket has
11    been effectively paired.
12    """
```

18. In a few words, what is the task this function aims to implement? *

19. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

20. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

21. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 6:

Below are instructions for implementing the function:

Context

```
1 class Words2Numbers:
2     """
3     The class provides a text-to-number conversion utility, allowing conversion of
4     written numbers (in words) to their numerical representation.
5     """
6
7     def __init__(self):
8         """
9         Initialize the word lists and dictionaries required for conversion
10        """
11        self.numwords = {}
12        self.units = [
13            "zero", "one", "two", "three", "four", "five", "six", "seven", "eight",
14            "nine", "ten", "eleven", "twelve", "thirteen", "fourteen", "fifteen",
15            "sixteen", "seventeen", "eighteen", "nineteen",
16        ]
17        self.tens = ["", "", "twenty", "thirty", "forty", "fifty", "sixty",
18                    "seventy", "eighty", "ninety"]
19        self.scales = ["hundred", "thousand", "million", "billion", "trillion"]
20
21        self.numwords["and"] = (1, 0)
22        for idx, word in enumerate(self.units):
23            self.numwords[word] = (1, idx)
24        for idx, word in enumerate(self.tens):
25            self.numwords[word] = (1, idx * 10)
26        for idx, word in enumerate(self.scales):
27            self.numwords[word] = (10 ** (idx * 3 or 2), 0)
28
29        self.ordinal_words = {'first': 1, 'second': 2, 'third': 3, 'fifth': 5,
30                              'eighth': 8, 'ninth': 9, 'twelfth': 12}
31        self.ordinal_endings = [('ieth', 'y'), ('th', '')]
32
33
34    def text2int(self, textnum):
35        pass
36
37
38
```

Instruction 1

```
1 def is_valid_input(self, textnum):
2     """
3     Check if the input text ""textnum"" contains only valid words that
4     can be converted into numbers. Return ""True"" if the input is valid,
5     otherwise return ""False"". Initially, the function modifies the input
6     text by replacing hyphens with spaces using ""textnum.replace('-', ' ')""
7     to standardize the format. It then iterates over each word in the modified
8     text using a for loop. If a word matches items in the class attribute
9     ""self.ordinal_words"", it skips further checks for that word. For other words,
10    it checks for specific suffix patterns that indicate ordinal numbers, listed
11    in ""self.ordinal_endings"", and modifies them accordingly. If the processed
12    word does not exist in the comprehensive dictionary ""self.numwords"" of
13    recognized number words, the function returns ""False"". If all words are
14    recognized, it returns ""True"".
15    :param textnum: The input text containing words representing numbers.
16    :return: True if input is valid, False otherwise.
17    """
```

Instruction 2

```
1 def is_valid_input(self, textnum):
2     """
3     Check if the input text contains only valid words that can be converted
4     into numbers.
5     :param textnum: The input text containing words representing numbers.
6     :return: True if input is valid, False otherwise.
7     >>> w2n = Words2Numbers()
8     >>> w2n.is_valid_input("thirty-two")
9     False
10    """
```

Instruction 3

```
1 def is_valid_input(self, textnum):
2     """
3     Confirm whether the input string 'textnum' is composed only of words
4     convertible to numeric values. Return 'True' if it is, 'False' if not.
5     :param textnum: The input text containing words representing numbers.
6     :return: True if input is valid, False otherwise.
7     """
```

22. In a few words, what is the task this function aims to implement? *

23. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

24. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

25. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 7:

Below are instructions for implementing the function:

Instruction 1

```
1 def count_distinct_characters(string: str) -> int:
2     """
3     Create a function 'count_distinct_characters' that receives
4     'string' as its only argument. The purpose of this function
5     is to calculate and return the number of unique characters
6     in 'string', without considering uppercase and lowercase
7     distinctions. It begins by transforming 'string' to all lowercase
8     letters using 'string.lower()' for case insensitivity. Following this,
9     it changes the lowercase string into a set via 'set()', which
10    removes all duplicate characters. The function finally returns
11    the size of this set, counted by 'len()', indicating the count
12    of distinct characters from 'string'.
13    """
```

Instruction 2

```
1 def count_distinct_characters(string: str) -> int:
2     """
3     Given a string, find out how many distinct characters
4     (regardless of case) does it consist of
5     >>> count_distinct_characters('xyzXYZ')
6     3
7     >>> count_distinct_characters('Jerry')
8     4
9     """
```

Instruction 3

```
1 def count_distinct_characters(string: str) -> int:
2     """
3     Compose a function named 'count_distinct_characters'
4     which receives a single string parameter. The goal of
5     the function is to count the unique characters found
6     in the string, ignoring differences between uppercase
7     and lowercase letters. To accomplish this, the string
8     is converted to lowercase to standardize character comparison.
9     Subsequently, the string is turned into a set to discard duplicates.
10    Finally, the function reports the number of distinct characters
11    by returning the size of the set.
12    """
```

26. In a few words, what is the task this function aims to implement? *

27. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

28. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

29. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 8:

Below are instructions for implementing the function:

Context

```
1 class IPAddress:
2     """
3     This is a class to process IP Address, including validating,
4     getting the octets and obtaining the binary representation
5     of a valid IP address.
6     """
7
8     def __init__(self, ip_address):
9         """
10        Initialize the IP address to the specified address
11        :param ip_address:string
12        """
13        self.ip_address = ip_address
14
15
16    def is_valid(self):
17        pass
18
19    def get_binary(self):
20        pass
21
22
23
```

Instruction 1

```
1 def get_octets(self):
2     """
3     Deliver a list of octets of a valid IP address,
4     or return an empty list if the IP is not valid.
5     :return: list
6     """
```

Instruction 2

```
1 def get_octets(self):
2     """
3     If the IP address is valid, the list of four decimal
4     numbers separated by "." constituting the IP address
5     is returned; otherwise, an empty list is returned
6     :return: list
7     >>> ipaddress = IPAddress("10.10.10.10")
8     >>> ipaddress.get_octets()
9     ["10", "10", "10", "10"]
10    """
```

Instruction 3

```
1 def get_octets(self):
2     """
3     The function must first validate the IP address;
4     if it's valid, it splits the address into octets
5     using the dot ('.') separator and returns these o
6     ctets as a list. If the IP address does not pass
7     validation, an empty list is returned. It uses a
8     validation method to check the IP address's
9     validity before splitting.
10    :return: list"
11    """
```

30. In a few words, what is the task this function aims to implement? *

31. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

☐ Very Easy

☐ Easy

☐ Not Easy Nor Difficult

☐ Difficult

☐ Very Difficult

32. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

33. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 9:

Below are instructions for implementing the function:

Instruction 1

```
1 def right_angle_triangle(a, b, c):
2     """
3     Define the function 'right_angle_triangle', which receives three integer arguments:
4     'a', 'b', 'c', representing the lengths of triangle sides. The function should determine
5     whether these sides can create a right-angled triangle by assessing whether the sum of
6     the squares of any two sides is the square of the remaining side. It performs evaluations
7     using 'a ** 2 + b ** 2 == c ** 2', 'a ** 2 + c ** 2 == b ** 2', and 'b ** 2 + c ** 2 == a ** 2'
8     through a logical OR operation. It outputs True if any of the conditions are true, otherwise False.
9     """
```

Instruction 2

```
1 def right_angle_triangle(a, b, c):
2     """
3     Given the lengths of the three sides of a triangle.
4     Return True if the three sides form a right-angled triangle,
5     False otherwise. A right-angled triangle is a triangle in
6     which one angle is right angle or 90 degree.
7     Example:
8     right_angle_triangle(3, 4, 5) == True
9     right_angle_triangle(1, 2, 3) == False
10    """
```

Instruction 3

```
1 def right_angle_triangle(a, b, c):
2     """
3     Implement the function 'right_angle_triangle' which accepts three values corresponding
4     to the sides of a potential triangle. Applying the Pythagorean Theorem, the function
5     checks all side combinations to see if the square of one equals the sum of the squares
6     of the others, signifying a right-angled triangle. It returns True for a correct set,
7     otherwise False.
8     """
```

34. In a few words, what is the task this function aims to implement? *

35. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

36. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

37. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 10:

Below are instructions for implementing the function:

Context

```
1 import csv
2 class CSVProcessor:
3     """
4     This is a class for processing CSV files, including reading and writing CSV data,
5     as well as processing specific operations and saving as a new CSV file.
6     """
7
8     def __init__(self):
9         pass
10
11     def read_csv(self, file_name):
12         pass
13
14     def write_csv(self, data, file_name):
15         pass
16
17
18
```

Instruction 1

```
1 def process_csv_data(self, N, save_file_name):
2     """
3     Retrieve data from a csv file identified as 'save_file_name',
4     pick out the column at position N from the data, ensure all of
5     its components are in uppercase, and then lodge the altered
6     content into a new csv file, maintaining the primary titles.
7     The new file's name will incorporate '_process' right before
8     the extension of 'save_file_name'. Initially employ 'read_csv'
9     to pull titles and data from 'save_file_name', then operate on
10    extracting and capitalizing the N-th column, finishing by
11    preserving these in a new csv file using 'write_csv' and
12    amending the file name suitably. Return the output of 'write_csv'.
13    :param N: int, the N th column(from 0)
14    :param save_file_name, the name of file that needs to be processed.
15    :return:int, if success return 1, or 0 otherwise
16    """
```

Instruction 2

```
1 def process_csv_data(self, N, save_file_name):
2     """
3     Read a csv file into variable title and data.
4     Only remain the N th (from 0) column data and Capitalize them,
5     store the title and new data into a new csv file.
6     Add '_process' suffix after old file name, as a new file name.
7     :param N: int, the N th column(from 0)
8     :param save_file_name, the name of file that needs to be processed.
9     :return:int, if success return 1, or 0 otherwise
10    >>> csvProcessor = CSVProcessor()
11    >>> csvProcessor.read_csv('read_test.csv')
12    (['a', 'b', 'c', 'd'], [['hElLo', 'YoU', 'ME', 'LoW']])
13    >>> csvProcessor.process_csv_data(0, 'read_test.csv')
14    1
15    >>> csvProcessor.read_csv('read_test_process.csv')
16    (['a', 'b', 'c', 'd'], [['HELLO']])
17    """
```

Instruction 3

```
1 def process_csv_data(self, N, save_file_name):
2     """
3     Read a csv file named ""save_file_name"", extract only the N-th column
4     from the data, capitalize all its elements, and store the result with
5     the original titles into a new csv file. The new file name should be
6     the original ""save_file_name"" with a ""_process"" suffix added before
7     the file extension. The function begins by calling the ""read_csv""
8     method with ""save_file_name"" to retrieve the data and titles. It then
9     extracts the N-th column from the data using list comprehension with
10    ""row[N]"". Each element of this column is transformed to uppercase using
11    another list comprehension. This new column along with the original titles
12    is saved into a new csv file by calling the ""write_csv"" method, where
13    the file name is generated by appending '_process.csv' to the prefix of
14    ""save_file_name"". Return the output of the method ""write_csv"".
15    :param N: int, the N th column(from 0)
16    :param save_file_name, the name of file that needs to be processed.
17    :return:int, if success return 1, or 0 otherwise
18    """
```

38. In a few words, what is the task this function aims to implement? *

39. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

40. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

41. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 11:

Below are instructions for implementing the function

Instruction 1

```
1 def f(n):
2     """
3     Write a method called 'f' to accept an integer 'n' and formulate
4     a list of 'n' integers. Corresponding to each list slot, designated
5     by 1-based index 'i', even index values calculate to the factorial
6     of 'i' (complete product from 1 to 'i'), whereas odd index values
7     compute to the total sum from 1 to 'i'. This method sets up some
8     initial values for the cases where 'n' lower than 2 and then continues
9     to populate the list by deciding, based on the index's oddity or evenness,
10    whether to multiply for factorials or add for sums.
11    """
```

Instruction 2

```
1 def f(n):
2     """
3     Implement the function f that takes n as a parameter,
4     and returns a list of size n, such that the value of
5     the element at index i is the factorial of i if i is even
6     or the sum of numbers from 1 to i otherwise.
7     i starts from 1.
8     the factorial of i is the multiplication of the numbers
9     from 1 to i (1 * 2 * ... * i).
10    Example:
11    f(5) == [1, 2, 6, 24, 15]
12    """
```

Instruction 3

```
1 def f(n):
2     """
3     Develop a function 'f' that accepts an integer 'n' and returns a list
4     of 'n' integers. The list is formed by calculating factorial for even
5     indices 'i' and the summation of numbers up to 'i' for odd indices,
6     starting from 1.
7     """
```

42. In a few words, what is the task this function aims to implement? *

43. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

44. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

45. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 12:

Below are instructions for implementing the function:

Context

```
1 class VendingMachine:
2     """
3     This is a class to simulate a vending machine, including adding products,
4     inserting coins, purchasing products, viewing balance, replenishing product
5     inventory, and displaying product information.
6     """
7
8     def __init__(self):
9         """
10        Initializes the vending machine's inventory and balance.
11        """
12        self.inventory = {}
13        self.balance = 0
14
15    def add_item(self, item_name, price, quantity):
16        pass
17
18    def insert_coin(self, amount):
19        pass
20
21    def restock_item(self, item_name, quantity):
22        pass
23
24    def display_items(self):
25        pass
26
27
28
```

Instruction 1

```
1 def purchase_item(self, item_name):
2     """
3     Initiate a purchase of an item referenced by 'item_name' from the vending
4     machine, update and return the balance if the item is bought. Otherwise,
5     return 'False' if the item is not in stock or the balance isn't enough.
6     :param item_name: The name of the product to be purchased, str.
7     :return: If successful, returns the balance of the vending machine after
8     the product is purchased, float, otherwise, returns False.
9     """
```

Instruction 2

```
1 def purchase_item(self, item_name):
2     """
3     Purchases a product from the vending machine and returns the balance
4     after the purchase and display purchase unsuccessful if the product
5     is out of stock.
6     :param item_name: The name of the product to be purchased, str.
7     :return: If successful, returns the balance of the vending machine
8     after the product is purchased, float,otherwise,returns False.
9     >>> vendingMachine = VendingMachine()
10    >>> vendingMachine.inventory = {'Coke': {'price': 1.25, 'quantity': 10}}
11    >>> vendingMachine.balance = 1.25
12    >>> vendingMachine.purchase_item('Coke')
13    0.0
14    >>> vendingMachine.purchase_item('Pizza')
15    False
16    """
```

Instruction 3

```
1 def purchase_item(self, item_name):
2     """
3     Purchase a product specified by ""item_name"" from the vending machine
4     and update the balance accordingly. Return the updated balance if the
5     purchase is successful, otherwise return ""False"" if the product is
6     out of stock or insufficient balance.
7     :param item_name: The name of the product to be purchased, str.
8     :return: If successful, returns the balance of the vending machine
9     after the product is purchased, float,otherwise,returns False.
10    """
```

46. In a few words, what is the task this function aims to implement? *

47. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

☐ Very Easy

☐ Easy

☐ Not Easy Nor Difficult

☐ Difficult

☐ Very Difficult

48. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

49. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Demographics Information

50. What is your highest education qualification? *

Une seule réponse possible.

- ☐ Ph.D.
- ☐ Master's Degree
- ☐ Bachelor Degree
- ☐ Autre :

51. What is your field of work? (Engineer, Researcher, Developer, etc.) *

52. How many years of development experience do you have? *

Une seule réponse possible.

- ☐ Less than 5
- ☐ Between 5 and 10
- ☐ More than 10

53. Thank you for completing the survey! Any feedback you would like to provide?

Google Forms

